

Two-dimensional and three-dimensional multiple object tracking learning performance in adolescent female soccer players: The role of flow experience reflected by heart rate variability

Xiang Che^{a,c}, Yu Zhang^b, Jingkang Lin^{a,c}, Kun Zhang^d, Weiqun Yao^{a,c}, Jijun Lan^{a,c,**}, Jie Li^{e,f,*}

^a School of Psychology, Shaanxi Normal University, Xi'an, China

^b School of Psychology, Beijing Sport University, Beijing, China

^c Key Laboratory of Behavior and Cognitive Neuroscience of Shaanxi, Shaanxi Normal University, Xi'an, China

^d Physical Education Institute, Shaanxi Normal University, Xi'an, China

^e Center for Cognition and Brain Disorders, the Affiliated Hospital, Hangzhou Normal University, Hangzhou, China

^f Institutes of Psychological Sciences, Hangzhou Normal University, Hangzhou, China

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ABSTRACT

Three-dimensional multiple object tracking (3D-MOT) has been used in various fields to mimic real-life tracking, especially in perceptual-cognitive skills training for soccer. Yet, the learning efficiency in 3D-MOT tasks has not been compared with 2D-MOT. Further, whether the advantage can be reflected by heart rate variability (HRV) based on the neurovisceral integration model should also be examined. Therefore, we used both 2D- and 3D-MOT in a brief adaptive task procedure for adolescent female soccer players with HRV measurement. A faster tracking speed threshold of participants was found in the 3D- compared to 2D-MOT, as well as average tracking speed in the last training period of 3D-MOT. Moreover, lower low frequency (LF) components of HRV in the 3D-MOT indicated a flow experience, demonstrating the provision of more attentional resources. Therefore, we observed that adolescent female soccer players demonstrated higher learning efficiency in 3D-MOT tasks in virtual reality (VR) through a higher flow experience. This study examined the learning efficiency between the two MOT tasks in the soccer domain using evidence from HRV and highlighted the utility and applicability of 3D-MOT application.

1. Introduction

In a variety of everyday situations, people can observe numerous objects moving in their vicinity, such as other people, vehicles, or animals. We track these multiple moving objects for various purposes. For instance, as drivers maneuver their vehicles across busy city intersections, they need to keep track of other moving vehicles as well as pedestrians walking on the street to maintain safety and avoid accidents. This cognitive ability is known as multiple object tracking (MOT), a concept first introduced by Pylyshyn and Storm [60] to determine how people track multiple objects in their environment.

1.1. Two-dimensional (2D) and three-dimensional MOT (3D-MOT)

The classical MOT tasks were presented using 2D. In 2D-MOT tasks, identical objects were randomly arrayed, and observers tracked a fixed number of targets among visually similar distractors. MOT capacity is generally limited to four objects [60,70]. However, there is another MOT task in which stimuli are presented with 3D graphic structures and require binocular vision to track. The task more closely depicts a 3D scene in real-life situations. We thus define this MOT task as a 3D-MOT task [18,52].

The main difference between 2D- and 3D-MOT is the additional information provided by the 3D scene. While in 3D environments, people may perceive additional information as follows: visual structure

* Corresponding author at: Room 119, No. 18 Shuyuan Building, Cangqian Campus, Hangzhou Normal University, Hangzhou, 311121, China.

** Corresponding author at: Room 9407, No. 9 Teaching Building, 199 South Chang'an Road, Yanta Campus, Shaanxi Normal University, Xi'an, 710062, China.

E-mail addresses: xiang.che@snnu.edu.cn (X. Che), xingzhilab@126.com, lanjijun@snnu.edu.cn (J. Lan), lijie.psy@hznu.edu.cn (J. Li).

shadings (reflecting different rays from the environment) [32,40], object occlusions with binocular disparity [9], and visual sizes (near-large, far-small) [61]. The binocular vision disparity, as demonstrated by the performance differences in 2D and 3D scenes, provides a visual cue related to the object and environment [12]. However, visual structure shadings and occlusion are more complex for 3D stimuli than for 2D tasks [9,56].

1.2. 3D-MOT and flow experience

Virtual reality (VR) is one environment in which 3D-MOT can be presented [34]. A VR environment provides a more realistic 3D scene for 3D-MOT and a more user-friendly experience compared to using a computer [93]. Previous studies indicate that the flow induced by more realistic 3D scenes in VR encourages people to participate in this environment [8,10,46].

Moreover, the adaptive procedure (also called the staircase procedure, see [18]) has been widely utilized in 3D-MOT tasks. The motion speed of each trial corresponds to participant tracking ability (i.e., correctly tracking all objects, speed increased; errors made, speed decreased). Because variable motion speed is used and the observer needs to dynamically process the information, the demands on their perceptual-cognitive functions vary [18]. Motion speed can be an indicator of attentional demands in MOT tasks [82]. This adaptive procedure is in line with the “challenge-skill balance” proposed by Csikszentmihalyi [13], one of the essential components to induce flow experience, which indicates a person’s skill is at just the right level to cope with the situation demands.

The flow experience is a state of profound task absorption, cognitive efficiency, and intrinsic enjoyment that makes an individual experiencing flow feel immersed in an activity [45]. Flow experience occurs in a state of moderate physiological arousal and can thus be measured by heart rate variability (HRV) [53]. HRV is based on continuous electrocardiogram (ECG) recording [73,92]. The low frequency (LF) component of HRV is a better indicator of flow than other indicators [83]. Previous studies indicated that a VR environment can provide immersion and realism, which also induces the flow experience leading to HRV decrease [8,91]. The fit (skills matching demands) condition of a task can also induce HRV decrease and reflect flow [29,83].

The adaptive procedure in an MOT task matches the participants’ abilities. Additionally, feedback is provided at the end of each MOT trial. Therefore, the adaptive procedure can be regarded as a learning process, and its performance can be regarded as learning efficiency [11,62]. Previous studies suggested that tasks performed in a VR environment thereby promote learning [25]. However, the promoted learning efficiency may result in a flow experience. Bodzin et al. [10] adopted an immersive virtual reality learning game for high school students. The entire study sample ($n = 57$) reported a flow experience when they played the learning game. Moreover, learners who experience flow in VR can improve their learning efficiency and alter their attitudes and behavior, becoming more engaged in taking constructive action [16]. Further, for learning in MOT, flow experience may induce prefrontal cortex (PFC) activity [67] and mobilize attention resources, leading to a positive learning effect [21]. However, PFC activity can be reflected by HRV [79,81], and PFC activity is also associated with MOT tasks [44], especially with motion speed [71].

In summary, flow experience provided a positive effect on learning and cognitive ability. Further, flow is associated with 3D scenes in the VR environment and adaptive MOT procedures. Therefore, we hypothesize that flow experience plays an important role in 3D-MOT under the adaptive procedure [8,10,29]. For comparison of induced flow for 3D-MOT learning, we needed to construct a 2D-MOT that corresponded with a 3D-MOT to investigate the effect of flow difference between 2D- and 3D-MOT.

1.3. Effect on perceptual-cognitive skills in soccer

3D-MOT has been widely applied, especially in soccer, owing to its realism. In soccer games, a player is required to track other individual players on the field, for example, knowing where the team’s forwards are currently located and passing the ball to complete a successful assist or knowing where opposing players are located to defend effectively. This cognitive processing is regarded as part of the perceptual-cognitive skills in soccer [42]. Perceptual-cognitive skill is used in sports-specific context [86]. Mann et al. [42] indicated that the more realistic the perceptual-cognitive skill paradigm, the greater the measurement sensitivity. The 3D-MOT task is similar to a real soccer game situation [18]. Therefore, 3D-MOT can be trained to improve soccer players’ cognitive functions and game performance [18,35,68].

The 3D-MOT has already been confirmed as effective in training and evaluating aspects of sport performance, such as improved motor-response decision-making for badminton players [66] and decision-making accuracy when soccer players passing balls [54]. In addition, flow experience as a positive mental state can contribute to athletes’ game performance [75] and reduce errors during a game [28]. Long-term flow perception can contribute to maintaining focus and avoiding disruptive factors such as negative thoughts or performance mistakes, and thus keep athletes performing at their peak for longer [76]. So far, it remains unclear whether the improvement of soccer performance by 3D-MOT training is associated with the perceived flow experience during the 3D-MOT task. Therefore, investigating and examining the flow experience in 3D-MOT can enhance our understanding of 3D-MOT and, thus, improve its utility and applicability for soccer.

Adolescent soccer players may be appropriate for this investigation. Their perceptual-cognitive skills are improving significantly [88] and they are in the process of developing a high level of cognitive abilities, including MOT [6]. Investigating the learning difference between 2D- and 3D-MOT would provide additional guidance on how to select training tools. Importantly, Huijgen et al. [26] indicated that the age of approximately 15 years is an essential age for soccer players because they may be offered their first professional contracts at this age. Therefore, our research on perceptual-cognitive skills for adolescent female soccer players can enrich the field and contribute to promoting further development of the sport.

1.4. The present study

3D-MOT tasks as a tool for training perceptual-cognitive skills have been widely used among soccer players [18,35]. 3D-MOT tasks may approximate real soccer play more closely than 2D tasks do. Mann et al. [42] indicate that the more realistic the perceptual-cognitive skill paradigm, the greater the measurement sensitivity. Further, adolescence is a key time period for perceptual-cognitive skill development and future career decision-making for soccer players. So far, the learning efficiency, which is the basis of training [41], has not been compared between 3D-MOT and 2D-MOT. Investigating learning efficiency could address the utility and applicability of 3D-MOT. Therefore, we constructed a 2D-MOT that corresponds with 3D-MOT to investigate the learning efficiency difference for adolescent soccer players.

3D-MOT tasks are typically performed in a VR environment under adaptive procedures, which may induce a greater flow experience. Flow experience can contribute to learning, and research has shown that flow experience can be reflected by HRV. So far, it remains unclear yet whether the flow experience is associated with the learning efficiency in 3D-MOT tasks. Investigating the mechanism can enhance the understanding of 3D-MOT, thus improving its utility and applicability in soccer. Therefore, in the present study, we aimed to measure the HRV of adolescent soccer players during MOT tasks to investigate the association of flow experience with learning efficiency in 3D-MOT.

2. Materials and methods

2.1. Participants

In this study, we sought to recruit adolescent female players from a provincial U15 soccer team through coaching staff. The inclusion criteria for participants in this study were as follows: The participants must be healthy adolescents of the female sex who are approximately 13–18 years old and have normal vision; participants must participate in soccer training sessions for, on average, more than 15 h per week and have no previous experiences with MOT tasks. In addition, their ECG measurement must be successful to calculate valid HRV parameters [47]. However, any such potential participant would be excluded if their artifact time of ECG were to exceed 5% of the recorded time in this study [90]. Each participant would be required to sign a written informed consent before participating in the study. This study was approved by the Ethics Committee of Shaanxi Normal University.

2.2. Apparatus, stimuli, and procedure

The 3D and 2D-MOT tasks were both created using the Unity™ engine (Unity Technologies, United States, <https://www.unity.com>). The 3D mode was presented in the SteamVR™ v1.2 program environment (Valve Corporation, United States, <https://www.steamvr.com>). A cross-over design was used in this study.

2.2.1. 3D-MOT task

All 3D-MOT trials were performed in a VR environment via a kit from HTC Vive™ (HTC Corporation, Taiwan, China, <http://www.vive.com>), including HTC Vive Pro-head-mounted display (HMD), HTC Vive Controller and SteamVR™ Base Stations 2.0. The HMD had a combined resolution of 2880×1600 (1440×1600 for a single eye). Its refresh rate was 90 Hz. The field of view was 110° , and the pupil spacing could be adjusted for each subject. Participants engaged in the 3D-MOT task using the HMD via HTC Vive Controller, and the location information was tracked by two base stations (SteamVR™ Base Stations 2.0).

In the HMD VR environment, we designed a virtual rectangular-shaped room with a length of 5 m, a width of 5 m, and a height of 2.5 m. The room had white walls and a blue floor. The HMD was located 4 m away from the nearest boundary of the room. The stimulus objects (yellow balls) moved randomly in the virtual room space. The vertical cross-section (5 m wide and 2.5 m in height) was defined along the midline of the length of the room. The distance from participants to this cross-section was 6.5 m. The cross-section had an identical horizontal visual angle (42.07°) and vertical visual angle (21.77°) to that in the 2D-MOT task. Each ball had a diameter of 27.94 cm (2.46° average) and an identical initial motion speed of 0.68 m/s (5.97° /s).

We utilized a 3D-MOT task (see Fig. 1), which has been developed, refined, and widely evaluated by several researchers [17,22,65]. All

participants sat in the valid control area of the two base stations in a quiet room. An HMD was fixed on the chair of the room. The HMD had controls for moving within the virtual environment described. The experimenter instructed and assisted participants in placing the HMD and adjusting the pupil spacing to maintain the clarity of the HMD visual display.

Next, we presented a practice block of five trials to ensure the participant understood the task. At the beginning of each trial, there were 10 yellow balls located randomly in the virtual room. After the participant pulled the trigger of the hand controller, four of the 10 yellow balls began to flicker in a red color (2 s.), which indicated these objects should be tracked. Then, the four balls changed to yellow, and a total of 10 balls began randomly moving in the virtual room for 8 s. While moving, the objects randomly changed direction when they hit the edge of the space or collided with each other. Participants were instructed to track the four marked targets during the entire period. Finally, all of the balls stopped moving. The participants were asked to choose and click the four balls that they had tracked using the hand controller. Feedback for the four correct objects was indicated by a purple-to-yellow color change until the beginning of the next trial. The experiment was based on an adaptive staircase procedure [33,54]. If participants correctly tracked all four balls, the motion speed increased by 30% over the speed in the previous trial. However, if participants made errors, the motion speed decreased by 20%.

2.2.2. 2D-MOT tasks

The 2D-MOT trials were performed on a 21.5-inch desktop display monitor with a resolution of 1920×1080 . Participants were directed to maintain a distance from head to monitor of 60 cm to guarantee similarity of the visual angles of objects, stimuli area, and moving speed in the 3D task.

The 2D-MOT task procedure was identical to the 3D-MOT task (see Fig. 1). All participants completed five practice trials and 30 trials of the 2D-MOT task sitting in front of the monitor. The experiment was based on the same staircase procedure as that in the 3D task. Participants were instructed to press the SPACEBAR of the keyboard to start each trial and use a mouse to choose and click the correct objects. Ten yellow discs were randomly distributed on the blue background of the monitor. Each disk had a diameter of 2.46° and an initial motion speed of 5.97° /s. The display area of the monitor was 42.18 cm long (42.07°) by 21.09 cm wide (21.77°).

2.3. HRV recording

Each participant's physiological data were recorded while they performed both the 2D and 3D-MOT tasks. We adopted SOMNotouch™ RESP (SOMNomedics, Germany, <http://www.somnomedics.de>), which is a multifunctional device that permits continuous and noninvasive electrocardiogram (ECG) recording. The device has three disposable Ag-

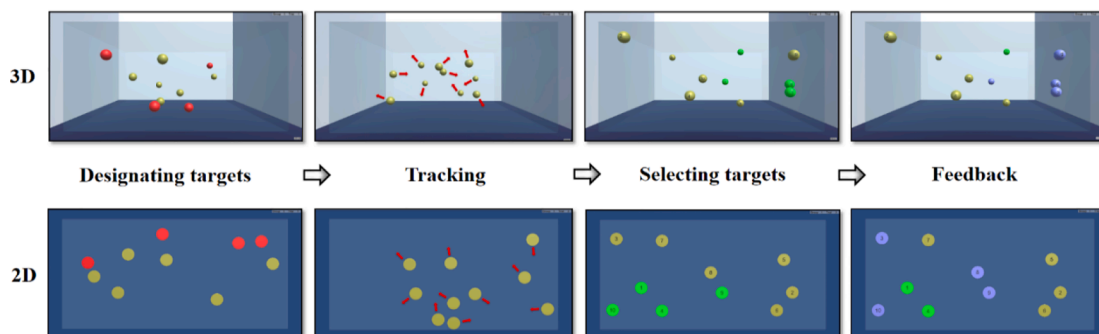


Fig. 1. MOT procedure in two modes. For both 3D and 2D modes, identical numbers of objects to be tracked and initial motion speed were presented, using a one-up one-down staircase procedure.

AgCl electrodes affixed on the right and left clavicles and the lower left rib of each participant. Subsequently, all three disposable electrodes transfer ECG data to the SOMNOtouch™ RESP's ECG sensor, which was fixed around the participants' torsos by a body belt during tasks.

ECG data were recorded at a sample bandwidth of 512 Hz. First, the data were input to DOMINO light software (Version 1.5.0, SOMNO-medics, Germany, <http://www.somnomedics.de>) to output participant R-R intervals. The length is the interval represented by the space between the R peaks of two consecutive heartbeats in the electrocardiogram (ECG) [89]. These R-R interval data were de-trended using the smoothness priors method with a parameter of 500. These data were analyzed using Kubios HRV standard software ver. 3.5.0 [77]. After analysis, time- and frequency-domain HRV parameters were calculated.

2.4. Outcomes

2.4.1. Learning efficiency of 2D- and 3D-MOT

All participants were instructed to complete a total of 30 trials in the 2D and 3D-MOT tasks. In this study, the variables of speed ratio threshold and average speed of time periods were analyzed as learning efficiency.

- (1) The last four inversions represent a stable indication of threshold, as adopted in various previous studies to evaluate stable, near-maximal MOT performance [17,84]. We calculated the speed ratio threshold from the average value of the last four inversion pairs of trials (i.e., four pairs of correct and incorrect trials) and average speed.
- (2) Speed change per trial is potentially informative, as it represents gradual performance change throughout the task. This value represents tracking speed information over time for 2D- and 3D-MOT tasks (see Fig. 3). We thereby divided all 30 trials into the first period (first 10), middle period (second 10), and last period (final 10), based on previous studies [37,38]. We also calculated the average tracking speed of three time periods in both the 2D and 3D-MOT tasks.

2.4.2. HRV measure

HRV analysis is based on temporal oscillations of successive heart-beat variations from a continuous ECG recording [73,78]. In this study, time-domain and frequency-domain variables were analyzed.

- (1) Time-domain HRV parameters, we calculated mean R-R standard deviations of normal-to-normal RR intervals (SDNN). These values reflect sympathetic and parasympathetic nervous activity. The root mean square of successive differences (RMSSD), a reflection of parasympathetic nervous activity, has been widely used for monitoring soccer training [63,64].
- (2) Frequency-domain analysis is another method for evaluating the balance of the sympathetic nervous system (SNS) and parasympathetic nervous system (PNS). It can provide more precise data in terms of direction and magnitude than can time-domain analysis [59]. For frequency-domain HRV parameters, we calculated the high frequency (HF) component, which ranges from 0.15 to 0.4 Hz and is mediated almost exclusively by parasympathetic nervous activity. The low frequency (LF) component, ranging from 0.04 to 0.15 Hz, is typically used to measure sympathetic efferent activity. The HF and LF data were also natural log-transformed. Relative powers of the LF and HF bands were assessed [77]. Moreover, the LF/HF ratio is an index of the relative sympathovagal balance [7].

2.5. Statistical analysis

Power analysis (G-Power, 3.1.9 version) calculated using the effect size ($f = 0.25$) indicated that a total sample of 24 participants would

yield 0.90 power at $\alpha = 0.05$. Data analyses were performed using the software IBM SPSS (Statistical Package for Social Sciences, version 26.0).

In both the 2D- and 3D-MOT tasks, all the data satisfied the normalized distribution from the results of the Shapiro-Wilk test ($p > 0.05$). Therefore, the within-subjects repeated-measures analysis of variance (ANOVA) was performed to analyze the difference, in terms of speed ratio threshold and speed ratio of the time period, between 2D-MOT and 3D-MOT.

For the HRV performance during the two MOT tasks, we found that some of the HRV parameters (SDNN, RMSSD, LF, HF) did not satisfy the normalized distribution from the results of the Shapiro-Wilk test ($p < 0.05$). Due to the skewed distribution of HRV recordings, parameters of SDNN, RMSSD, LF, and HF data were log-transformed by taking the natural logarithm (Ln) [57,58]. Therefore, we calculated the Ln SDNN, Ln RMSSD, Ln HF, and Ln LF parameters to satisfy the normalized distribution, then submitted them to all the HRV parameters into two-way within-subjects repeated-measures of analysis of variance (ANOVA).

3. Results

3.1. Participants

A total of 26 consenting participants were recruited and participated in both the 2D- and 3D-MOT tasks with the order randomly assigned. Specifically, 13 participants were first assigned to the 2D-MOT task then assigned to 3D-MOT, and the other 13 participants also assigned to both of the MOT tasks with an opposite order. However, two of the 26 participants' ECG recordings displayed discontinuity (over 5%) in some period. Given that the discontinuity could affect the precision of the HRV analyses [73,90,92], the two participants were excluded from the final analyses. Hence, data from 24 adolescent female soccer players from a provincial U15 soccer team (see Table 1) were analyzed in this study.

3.2. MOT performance

There was a significant main effect of Mode (2D, 3D), $F(1, 23) = 6.228, p = 0.020, \eta_p^2 = 0.213$, indicating a higher speed tracking ability in 3D than 2D mode (see Fig. 2).

Speed change percent per trial indicates different tracking speeds over the 30 trials. Subsequently, we performed a 2-Modes (2D, 3D) $\times$ 3-Time Periods (First, Middle, Last) within-subjects factors repeated-measures ANOVA. There was a significant main effect of time period, $F(2, 46) = 13.664, p < 0.001, \eta_p^2 = 0.373$. Post-hoc analysis with Bonferroni correction (used for all simple effect analyses here) indicated that the tracking speeds of the Last period ($M = 1.748 \pm 0.120, p < 0.001$) and Middle period ($M = 1.679 \pm 0.135, p = 0.006$) were significantly faster than that of the First period ($M = 1.358 \pm 0.079$). Importantly, there was a significant interaction between Mode and Time period, $F(2, 46) = 3.506, p = 0.038, \eta_p^2 = 0.132$. Post-hoc analysis indicated that

Table 1
Players' demographic information.

Age ($M \pm SD$, by years)	14.35 $\pm$ 0.42
Soccer Experience ($M \pm SD$, by years)	4.79 $\pm$ 1.25
Training per week ($M \pm SD$, by hours)	17.58 $\pm$ 1.24
N (Total)	24
Position of soccer players	
Center back	5
Center mid-fielder	2
Forward	2
Defensive mid-fielder	4
Goalkeeper	3
Side mid-fielder	3
Side back	5

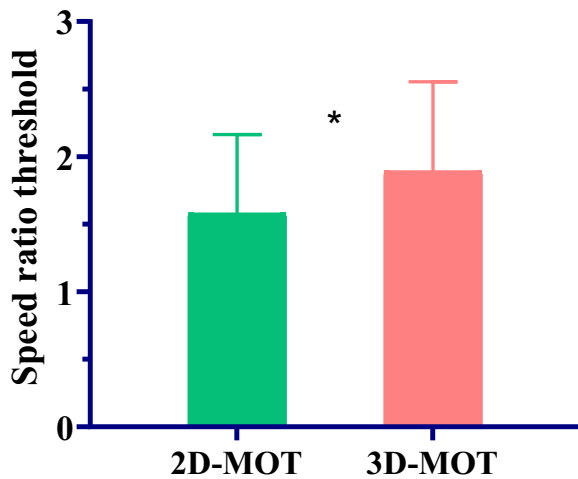


Fig. 2. Difference in the speed ratio threshold between 2D-MOT and 3D-MOT task for all participants ($M_{2D-MOT} = 1.581 \pm 0.582$, $M_{3D-MOT} = 1.894 \pm 0.660$, $p = 0.020$, $\eta_p^2 = 0.213$). The speed ratio is the change in objects' motion speed compared to its initial speed. Error bars denote the standard error of the mean (SEM). * $p < 0.05$.

adolescent female soccer players tracked 3D mode objects faster than they tracked 2D mode objects in the Last period ($M_{2D} = 1.603 \pm 0.125$, $M_{3D} = 1.893 \pm 0.141$, $p = 0.017$) but not in the First period ($M_{2D} = 1.319 \pm 0.096$, $M_{3D} = 1.396 \pm 0.099$, $p = 0.512$) or the Middle period ($M_{2D} = 1.695 \pm 0.153$, $M_{3D} = 1.663 \pm 0.141$, $p = 0.785$). In addition, adolescent female soccer players tracked objects faster in the Last period than the Middle period of 3D mode ($M_{Last} = 1.893 \pm 0.141$, $M_{Mid} = 1.663$, $p = 0.045$), compared to no significant increase between the Last period and the Middle period of 2D mode ($M_{Last} = 1.603 \pm 0.125$, $M_{Mid} = 1.695 \pm 0.153$, $p = 0.950$).

3.3. HRV

We also recorded and divided the R-R interval data of the three time periods. Then, we split these data into time- and frequency-domain HRV parameters (see Table 2). These data were also analyzed using a two-way within-subjects repeated-measures ANOVA (see Table 3). In the time-domain analysis, there were significant main effects of Mode on Ln SDNN ($F(2, 46) = 5.638$, $p = 0.026$, $\eta_p^2 = 0.197$). These results indicate

that SDNN in 3D mode was lower than that in 2D mode. In the frequency-domain analysis, there were significant main effects of mode on Ln LF ($F(2, 46) = 9.862$, $p = 0.005$, $\eta_p^2 = 0.300$). These values indicate that LF power in 3D mode was lower than that in 2D mode. None of the other effects was significant.

To investigate whether there was any proportional difference between each frequency, the percent occurrence of each HRV frequency was examined [55]. We separately submitted the percent of frequency (LF, HF) to a within-subjects repeated-measures ANOVA for both Modes and Time periods (First, Mid, Last). There was a significant main effect of percent of frequency in 2D mode, $F(1, 23) = 19.578$, $p < 0.001$, $\eta_p^2 = 0.460$, as well as in 3D mode, $F(1, 23) = 13.480$, $p = 0.001$, $\eta_p^2 = 0.370$. The main effect of time period was not significant in 2D mode, $F(2, 46) = 0.907$, $p = 0.411$, $\eta_p^2 = 0.038$, nor in 3D mode, $F(2, 46) = 2.195$, $p = 0.123$, $\eta_p^2 = 0.087$. The result indicated that the percentage of LF (the percent of LF power in total power) might be a dominant HRV index in MOT tasks (see Fig. 4); therefore, we inferred that LF plays a more important role in these MOT tasks.

4. Discussion

In the present study, we investigated the tracking speed of adolescent female soccer players in 2D- and 3D-MOT. The adolescent female soccer players displayed higher learning efficiency in the 3D-MOT task than in the 2D-MOT. From HRV analysis, this study found adolescent female soccer players performed a higher percentage of LF in both 2D and 3D-MOT, and especially lower LF component of HRV in 3D-MOT.

4.1. Learning efficiency of 2D and 3D-MOT

Tracking speed thresholds are reliable indicators of tracking ability, reflecting learning efficiency. In the one-up/one-down staircase procedure, the speed of the moving objects is gradually altered to match the tracking abilities of participants [33]. Higher speeds of moving objects disrupt participants' tracking, causing the participants to lose the tracked object or confuse the tracked object with a non-tracked object [96], resulting in decreased MOT performance.

Additionally, we considered speed change per trial as potentially informative, as it represents gradual performance change throughout the task. Therefore, we divided the average speed of all 30 trials into three time periods during the entire tracking procedure. Each MOT trial acted as a type of learning, as feedback was provided. We found that participants tracked 3D objects faster than they did 2D objects,

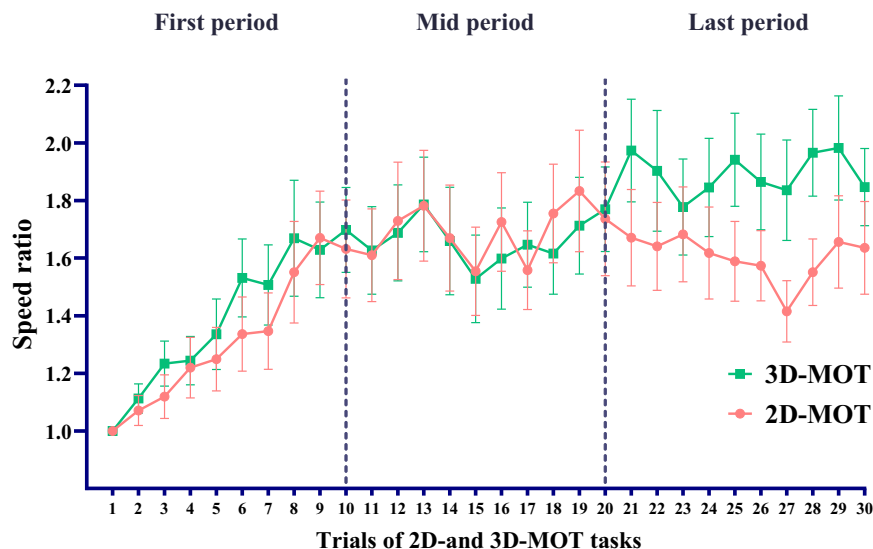


Fig. 3. Tracking speed differences per trial in 2D-MOT and 3D-MOT task. Error bars represent ± 1 SEM.

Table 2

Effects of the HRV on 2D and 3D MOT tasks in different time periods.

HRV	MOT mode	Time period ($M \pm SD$)			
		First	Mid	Last	Total
Time-domain					
Mean R-R (ms)	2D	736.758 $\pm$ 101.306	736.918 $\pm$ 88.7	737.567 $\pm$ 84.409	737.081 $\pm$ 90.451
	3D	730.128 $\pm$ 99.739	728.917 $\pm$ 97.576	733.422 $\pm$ 91.737	730.822 $\pm$ 95.061
Ln SDNN	2D	3.520 $\pm$ 0.423	3.561 $\pm$ 0.422	3.571 $\pm$ 0.466	3.551 $\pm$ 0.432
	3D	3.428 $\pm$ 0.443	3.362 $\pm$ 0.457	3.428 $\pm$ 0.374	3.414 $\pm$ 0.422
Ln RMSSD	2D	3.434 $\pm$ 0.566	3.522 $\pm$ 0.547	3.513 $\pm$ 0.568	3.489 $\pm$ 0.554
	3D	3.452 $\pm$ 0.610	3.352 $\pm$ 0.602	3.447 $\pm$ 0.500	3.417 $\pm$ 0.566
Frequency-domain					
Ln HF	2D	5.714 $\pm$ 1.245	5.759 $\pm$ 1.186	5.848 $\pm$ 1.166	5.774 $\pm$ 1.184
	3D	5.565 $\pm$ 1.208	5.38 $\pm$ 1.245	5.612 $\pm$ 1.101	5.519 $\pm$ 1.174
Ln LF	2D	6.252 $\pm$ 0.892	6.347 $\pm$ 0.832	6.358 $\pm$ 0.936	6.319 $\pm$ 0.876
	3D	5.981 $\pm$ 0.737	5.92 $\pm$ 0.804	6.102 $\pm$ 0.911	6.001 $\pm$ 0.812
Percent of HF (%)	2D	35.91 $\pm$ 17.39	34.78 $\pm$ 14.559	35.987 $\pm$ 11.529	35.559 $\pm$ 14.491
	3D	37.409 $\pm$ 15.42	35.047 $\pm$ 15.103	37.104 $\pm$ 15.409	36.52 $\pm$ 15.131
Percent of LF (%)	2D	56.645 $\pm$ 16.72	59.356 $\pm$ 13.857	58.507 $\pm$ 12.114	58.169 $\pm$ 14.198
	3D	53.935 $\pm$ 14.878	55.654 $\pm$ 13.381	56.559 $\pm$ 14.485	55.382 $\pm$ 14.103
Ln (LF/HF ratio)	2D	0.539 $\pm$ 0.873	0.588 $\pm$ 0.663	0.510 $\pm$ 0.531	0.546 $\pm$ 0.694
	3D	0.416 $\pm$ 0.735	0.540 $\pm$ 0.720	0.487 $\pm$ 0.770	0.482 $\pm$ 0.736

2D: two-dimensional-MOT; 3D: three-dimensional-MOT; Mean R-R: time between RR intervals; SDNN: standard deviation of RR intervals; Ln SDNN: SDNN transformed by natural logarithm; RMSSD: the square root of the mean of the squares of the successive differences of the RR intervals; Ln RMSSD: RMSSD transformed by natural logarithm; High frequency (HF) (ms^2): frequency power ranging from 0.15 to 0.40 Hz; Low frequency (LF) (ms^2): frequency power ranging from 0.04 to 0.15 Hz; Ln HF: HF transformed by natural logarithm; Ln LF: LF transformed by natural logarithm; Percent of HF (%): the percent of HF power in total power; Percent of LF (%): the percent of LF power in total power; Ln(LF/HF ratio): low frequency (LF) (ms^2) / high frequency (HF) (ms^2) ratio transformed by natural logarithm.

Table 3

ANOVA of HRV on 2D and 3D MOT tasks in different time periods.

HRV	Mode			Time period			Time period $\times$ Mode		
	F	p	η_p^2	F	p	η_p^2	F	p	η_p^2
Time-domain									
Mean R-R (ms)	0.372	0.548	0.016	0.462	0.633	0.02	0.159	0.853	0.007
Ln SDNN	5.638	0.026*	0.197	1.071	0.351	0.045	1.349	0.269	0.055
Ln RMSSD	0.860	0.363	0.036	0.707	0.498	0.03	2.101	0.134	0.084
Frequency-domain									
Ln HF	1.807	0.192	0.073	1.554	0.222	0.063	0.836	0.44	0.035
Ln LF	9.862	0.005**	0.300	0.911	0.409	0.038	0.892	0.417	0.037
Percent of HF (%)	0.140	0.712	0.006	0.622	0.541	0.026	0.058	0.944	0.003
Percent of LF (%)	1.254	0.274	0.052	0.864	0.428	0.036	0.126	0.882	0.005
Ln (LF/HF ratio)	0.239	0.630	0.010	0.530	0.592	0.023	0.197	0.822	0.008

* $p < 0.05$, ** $p < 0.01$.

Mean R-R: time between RR intervals; SDNN: standard deviation of RR intervals; Ln SDNN: SDNN transformed by natural logarithm; RMSSD: the square root of the mean of the squares of the successive differences of the RR intervals; Ln RMSSD: RMSSD transformed by natural logarithm; High frequency (HF) (ms^2): frequency power ranging from 0.15 to 0.40 Hz; Low frequency (LF) (ms^2): frequency power ranging from 0.04 to 0.15 Hz; Ln HF: HF transformed by natural logarithm; Ln LF: LF transformed by natural logarithm; Percent of HF (%): the percent of HF power in total power; Percent of LF (%): the percent of LF power in total power; Ln(LF/HF ratio): Low frequency (LF) (ms^2) / High frequency (HF) (ms^2) ratio transformed by natural logarithm.

especially during the last period of the trials. This result indicates that after 30 similar training trials in different modes, those who received such learning in a 3D VR environment exhibited greater improvement in tracking speed than those who experienced a 2D environment presented on a PC screen during this brief MOT training.

This result is novel in comparison to previous studies, which simulated a 3D scene on a flat screen or manipulated several 3D cues. For instance, Liu et al. [36] found that additional 3D frame structures make a small contribution to MOT accuracy. Moreover, Vidakovic and Zdravkovic [85] investigated several categories of depth cues. Surprisingly, MOT tasks with all depth cues did not perform better than those with no cues. Further, they found that only object contrast cues (big size, gray color, near participants; small size, black color, far from participants) rather than other cues contributed more to tracking performance. In this study, the 3D-MOT task presented in a VR environment provided more 3D cues than previous studies and maximum realism, suggesting that the so-called “all cues” condition in Vidakovic and Zdravkovic [85] might not provide sufficient cues for 3D perception.

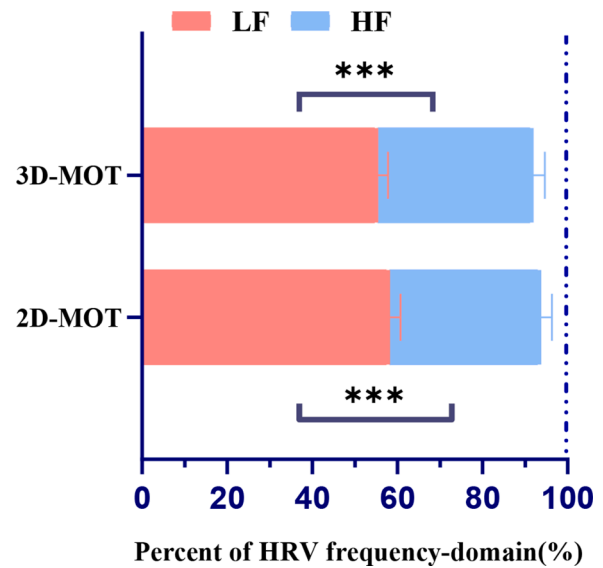


Fig. 4. Differences in the percentage of each HRV frequency domain in 2D and 3D-MOT ($M_{LF\ 2D-MOT} = 58.169 \pm 2.570\%$, $M_{HF\ 2D-MOT} = 41.831 \pm 2.598\%$, $p < 0.001$, $\eta_p^2 = 0.460$; $M_{LF\ 3D-MOT} = 55.382 \pm 2.420\%$, $M_{HF\ 3D-MOT} = 44.618 \pm 2.771\%$, $p < 0.001$, $\eta_p^2 = 0.460$). HF (ms^2): frequency power ranging from 0.15 to 0.40 Hz; LF (ms^2): frequency power ranging from 0.04 to 0.15 Hz. *** $p < 0.001$

Also, Liu et al. [36] and Vidakovic and Zdravkovic [85] have presented vague 3D cues effects from samples of normal participants. Previous studies indicated that experts performed better than normal in MOT tasks [15,95]. Soccer players are experts in 3D-MOT. Therefore, the study results also suggest that 3D-MOT as a tool for perceptual-cognitive skills training for soccer players results in better performance than classic 2D-MOT.

4.2. Flow experience reflected 3D-MOT learning

HRV analysis during 2D and 3D-MOT supports these results. First, we recorded all ECG data during 2D- and 3D-MOT tasks. Then, we converted these data into HRV parameters. Our results indicated that participants' HRV in 3D-MOT was lower than that in 2D-MOT in SDNN and LF domains from the time- and frequency-domain analysis. Furthermore, the percentage of LF was dominant compared to HF in all MOT tasks. Notably, we found that the decrease of the LF component in the 3D-MOT was more prominent than in the 2D-MOT. Previous studies indicated the LF component of HRV as an indicator of flow experience [23,43,83], which is in line with the present result.

In all MOT tasks, we found a higher percentage of LF. This may be due to the adaptive MOT procedures we adopted; using both the 2D and 3D-MOT tasks could have helped induce flow. The one-up/one-down adaptive procedure we adopted achieved equivalent relative task difficulty regardless of individual ability levels. The degree to which increasing task demands match cognitive ability may be essential in creating a flow experience [24]. Moreover, the procedure met the criteria for a flow experience; it offered a challenge-skill balance and clear feedback [13]. This theory was also supported by Keller et al. [29] and Tozman et al. [83], who highlighted flow induced by balanced skill-demand level (fit) reflected by LF-HRV, a better indicator than other indicators.

Further, the 3D-MOT induced more LF component of HRV decrease than 2D, indicating a higher flow experience during the task. Flow is negatively associated with the LF component: the more flow experienced, the more LF decreases [23]. 3D-MOT approaches real life and is performed in a VR environment, which can provide a high level of flow, engagement, and immersion [8,51]. Previous studies suggested that 3D scenes can induce more flow experience than 2D [48,74]. Also, the experience of flow in a VR environment may be more compelling than flow induced by tasks with desktop displays [51]. As indicated by positive effects in previous studies, flow in a VR environment is a mental state that accompanies complete involvement, enjoyment, and loss of the senses of time and space [72]. Another study found that LF changed during a VR task compared to HF, which remained stable [39,49]. Therefore, we regard 3D-MOT can provide more flow experience, reflected by lower LF-HRV than 2D-MOT.

However, apart from the discussion on what induced flow, how did flow contribute to MOT learning efficiency? Csikszentmihalyi [14] suggested flow experience is a balanced situation skill, provided there is a positive state model and facilitated concentration. He also proposed that flow can lead to learning by induction, intrinsic motivation, and autotelic activity. Performances in 3D scenes in VR environments resulted in better learning efficiency, which has always been associated with perceived flow during learning [10,16]. In addition, flow experience is also highly connected to the mobilization of attentional resources and activated in PFC [67], and thereby reflected in HRV due to the neurovisceral integration model [79,80]. However, the motion speed of MOT is also related to the activation of frontal eye fields (FEF) from the PFC region [44,71], which suggests the mobilization of attentional resources may also contribute to MOT performance. In summary, this evidence indicates more flow experience is provided by 3D-MOT with better learning efficiency versus 2D-MOT.

Furthermore, we found a lower SDNN value in the 3D-MOT than in the 2D. SDNN is a global index of HRV in this study. It reflects the overall (both short- and long-term) variation within the RR interval series [77].

Previous studies suggested that SDNN is correlated with LF power and that RMSSD correlates with HF power during sleep [50]. In addition, due to this underlying correlation, SDNN/RMSSD was confirmed to be a good surrogate for the LF/HF ratio in both patients and healthy controls [87]. Therefore, these studies provide evidence of lower SDNN and LF in 3D mode. These factors are indicators of combined sympathetic and parasympathetic nervous system activity. Thus, we can attribute the difference in SDNN to the correlation with LF power.

4.3. Flow experience to perceptual-cognitive skill in soccer

The present study confirmed the 3D-MOT learning efficiency contributed by inducing more flow experience. The result indicated that 3D-MOT is associated with more flow experience, highlighting its unique advantage in comparison to 2D-MOT. The additional perceived flow reflected by HRV may also contribute to sport match performance. Kim and Ko [30] suggested flow is also associated with shooting game performance for shoot players and can be reflected by HRV decreasing. Therefore, the positive effect of 3D-MOT training in soccer from previous studies may be partly attributed to its induced flow experience.

Tracking location and direction of teammates and opponents almost during all game time periods in soccer games. This cognitive processing is related to game performance which is associated with 3D-MOT [54, 68]. Flow experience for soccer players is team-based and is related to the teammates and opponents they perceive [5]. This study indicates the flow experience and 3D-MOT both occur simultaneously and are closely associated. Therefore, HRV changes as one of the indicators of flow experience should be measured during a 3D-MOT task to identify its effect. In team sports, HRVs are not unusually used as indicators for monitoring and evaluating mental state. HRV is highlighted as an important identification for soccer players such as fatigue [4,20,69] and anxiety [3]. In this study, we found that HRV can also serve as an indicator to measure the flow experiences that contribute to the performance of soccer players.

4.4. Limitations and future research

2D- and 3D-MOT differ in other ways that make them challenging to compare. For example, the motion patterns of 2D and 3D modes differ [31], leading to difficulties in judging speed. Task difficulty may decrease with low probability and frequency of collision and occlusion in 3D versus 2D tasks due to a large volume space in which objects are randomly moving [19,27,94]. However, apart from this evidence, the real-time vision angle changes of 3D object sizes may increase task difficulty [19].

This study adopted an adaptive staircase MOT procedure, in which tracking speed was a sensitive indicator of MOT ability. However, based on the model of the flexible attentional resource allocation in MOT, objects motion speed and tracking accuracy are indicators of trade-offs [1,2]. Future studies must utilize tracking accuracy as an indicator to consolidate the present result. In addition, longer time periods with more sessions were adopted in 3D-MOT learning for soccer players [54]. Future studies require measurement flow experience by HRV in 3D-MOT with more sessions over longer time periods to investigate its effect.

5. Conclusions

The present study found that adolescent female soccer players performed higher learning efficiency in 3D-MOT versus 2D-MOT tasks. This result was attributed to the positive effect of flow experience as evidenced by the lower LF and SDNN of HRV in the 3D-MOT versus 2D-MOT tasks. Further, this study highlights the role of flow experience in the utility and applicability of 3D-MOT in soccer sport.

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Declarations of Competing Interest

None.

Data Availability

Data will be made available on request.

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